# THE REDUCED SKEIN ALGEBRA OF THE ONCE-PUNCTURED BIGON AND THE SIMPLY-CONNECTED QUANTUM DOUBLE

Abstract. We give an explicit identification of the reduced stated $\mathrm{SL}_n$-skein algebra of the once-punctured bigon with the generalized quantum double of the two simply-connected quantum Borel algebras. The identification is specified on every horizontal matrix-coefficient arc. Its injectivity follows from an ordered two-arc factorization of the skein algebra, rather than from a comparison of defining relations alone. The prescribed diagonal quotient then identifies the two Cartan tori and gives the simply-connected quantized enveloping algebra.

## 1. Coefficients and conventions

Fix $n \geq 2$. We give the proof over

$$\mathbb{K} = \mathbb{Q}(v), \qquad q = v^{2n}, \qquad \lambda = q - q^{-1}.$$

Thus all fractional powers needed in the stated skein relations are available, and $\lambda$ is invertible. The same argument applies at non-root-of-unity specializations over a common ground field containing the required fractional powers. Throughout, $\mathcal{U}_q(\mathfrak{sl}_n)$ means the unrestricted, simply-connected quantum group, not a small quantum group.

We use the stated skein conventions of Lê–Sikora [3]: the product $xy$ places $x$ above $y$. The reduced algebra is the quotient by the bad corner arcs, as in [4, Section 7.1].

The surface $\mathfrak{B}^*$ is a disk with its top and bottom boundary points and one interior point removed. We fix the following labeling of the horizontal arcs. Both arcs point to the right; the first index is the left state and the second is the right state.

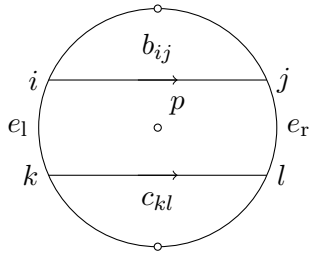



Figure 1. The labeling used in the formulas below.

At the upper corner the bad right-directed arcs are $b_{ij}$ with $i < j$. At the lower corner they are $c_{ij}$ with $i > j$. Consequently,

$$b_{ij} = 0 \quad (i < j), \qquad c_{ij} = 0 \quad (i > j) \qquad \text{in } \overline{\mathscr{S}}_n(\mathfrak{B}^*). \tag{1}$$

The reversal of the inequality at the lower corner is due to the 180° rotation identifying that corner with the standard upper-corner picture. If the two letters in a different drawing are interchanged, the formulas below are to be read with that interchange.

For the quantum group, put

$$\omega_{ai} = \delta_{a,i} - \delta_{a,i+1}, \qquad K_i = \widehat{K}_i \widehat{K}_{i+1}^{-1}.$$

The simply-connected presentation uses

$$\widehat{K}_a^{\pm 1} \quad (1 \le a \le n), \qquad E_i, F_i \quad (1 \le i < n),$$

with commuting toral generators and

$$\prod_{a=1}^{n} \widehat{K}_a = 1, \tag{2}$$

$$\widehat{K}_a E_i \widehat{K}_a^{-1} = q^{\omega_{ai}} E_i, \qquad \widehat{K}_a F_i \widehat{K}_a^{-1} = q^{-\omega_{ai}} F_i, \tag{3}$$

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{\lambda}. \tag{4}$$

For $X = E$ and $X = F$ the remaining relations are

$$X_i X_j = X_j X_i \qquad (|i-j| > 1), \tag{5}$$

$$X_i^2 X_j - (q + q^{-1}) X_i X_j X_i + X_j X_i^2 = 0 \qquad (|i-j| = 1). \tag{6}$$

Our coproduct convention is

$$\begin{aligned} \Delta(\widehat{K}_a) &= \widehat{K}_a \otimes \widehat{K}_a, \\ \Delta(E_i) &= E_i \otimes K_i + 1 \otimes E_i, \\ \Delta(F_i) &= F_i \otimes 1 + K_i^{-1} \otimes F_i. \end{aligned} \tag{7}$$

These conventions specify the extended quantum group in the question; see [2]. The corresponding Borel presentations use the toral relations and the respective $E$- or $F$-relations. They also follow from the quantum-matrix realizations recalled in Lemma 3.1 below.

## 2. The skein-theoretic matrix calculation

Let $V = \mathbb{K}^n$, let $e_{ab}$ denote its matrix units, and set

$$R = q^{-1/n} \left( \sum_{a,b=1}^{n} q^{\delta_{ab}} e_{aa} \otimes e_{bb} + \lambda \sum_{a>b} e_{ab} \otimes e_{ba} \right). \tag{8}$$

The braiding matrix is $PR$, where $P$ interchanges the tensor factors. This is the normalization used in [3, Equation (9)].

For a matrix $T = (t_{ij})$, write $T_1 = T \otimes I_n$ and $T_2 = I_n \otimes T$. The standard quantum coordinate algebra $\mathcal{O}_q(\mathrm{SL}_n)$ has presentation

$$R T_1 T_2 = T_2 T_1 R, \qquad \det_q(T) = 1, \tag{9}$$

where

$$\det_q(T) = \sum_{\sigma \in S_n} (-q)^{\ell(\sigma)} t_{1,\sigma(1)} \cdots t_{n,\sigma(n)}.$$

Define its triangular quotients

$$\mathcal{T}^- = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i < j), \qquad \mathcal{T}^+ = \mathcal{O}_q(\mathrm{SL}_n)/(t_{ij} : i > j). \tag{10}$$

**Lemma 2.1** (Two-arc matrix calculation)**.** *Let $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$, and let $B = (b_{ij})$ and $C = (c_{ij})$ be as in Figure 1.*

(1) *$B$ and $C$ satisfy* (9)*. Moreover, $B$ is lower triangular and $C$ is upper triangular.*

(2) *The mixed relation is*

$$RB_1C_2 = C_2B_1R. \tag{11}$$

*Equivalently, for all $i, j, k, l$,*

$$q^{\delta_{ik}} b_{ij} c_{kl} - q^{\delta_{jl}} c_{kl} b_{ij} = \lambda\big(\mathbf{1}_{j<l} c_{kj} b_{il} - \mathbf{1}_{k<i} b_{kj} c_{il}\big). \tag{12}$$

(3) *The homomorphisms from $\mathcal{T}^-$ and $\mathcal{T}^+$ determined by $T \mapsto B$ and $T \mapsto C$ are injective, and multiplication induces a vector-space isomorphism*

$$\mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ \xrightarrow{\sim} A, \qquad f \otimes g \longmapsto f(B)g(C). \tag{13}$$

*This is a vector-space assertion, not an assertion that the ordinary tensor-product multiplication is preserved.*

*Proof.* A thin neighborhood of either horizontal arc is an ideal bigon. The bigon theorem [3, Theorem 6.3] identifies its stated skein algebra with $\mathcal{O}_q(\mathrm{SL}_n)$, with the right-directed stated arcs as the matrix generators. This proves the quantum-matrix and determinant relations. Equation (1) gives the asserted triangularity.

Here are the boundary-exchange details for the mixed relation. They hold already in the unreduced algebra. Consider the two disjoint arcs with $c$ higher than $b$ at the left boundary and $b$ higher than $c$ at the right boundary. Denote the resulting mixed-height diagram, with external state pairs $(i, k)$ and $(j, l)$, by $W_{ik;jl}$. Applying the boundary height-exchange relation at the left or at the right gives, respectively,

$$W_{ik;jl} = \sum_{a,b} R_{ik,ab}\, b_{aj} c_{bl} = \sum_{a,b} c_{kb} b_{ia}\, R_{ab,jl}. \tag{14}$$

Here the flip in the braiding $PR$ has been removed by recording the indices in the order of the two arcs. Thus the coefficients appearing in the two exchanges are

$$R_{ik,ab} = q^{-1/n}\left(q^{\delta_{ik}}\delta_{ia}\delta_{kb} + \lambda \mathbf{1}_{k<i}\delta_{ka}\delta_{ib}\right),$$
$$R_{ab,jl} = q^{-1/n}\left(q^{\delta_{jl}}\delta_{aj}\delta_{bl} + \lambda \mathbf{1}_{j<l}\delta_{al}\delta_{bj}\right).$$

These are precisely the boundary braiding coefficients (8); the boundary relations and their braiding interpretation are those of [3, Section 4 and Theorem 6.4]. Substitution into (14), followed by cancellation of $q^{-1/n}$, proves (12).

It remains to prove the vector-space assertion, including the absence of additional relations after reduction. Write $S = \mathscr{S}_n(\mathfrak{B}^*)$ for the unreduced algebra, and temporarily use $b, c$ for the unreduced arcs. They form the standard saturated two-arc system for $\mathfrak{B}^*$: together with the boundary edges they give a groupoid spine of the punctured bigon. The ordered-arc factorization theorem, with the $b$-neighborhood above the $c$-neighborhood at both boundary edges, gives

$$m : \mathcal{O}_q(\mathrm{SL}_n) \otimes_{\mathbb{K}} \mathcal{O}_q(\mathrm{SL}_n) \xrightarrow{\sim} S, \qquad f \otimes g \longmapsto f(B)g(C). \tag{15}$$

The relevant result permits arbitrary orders at the boundary edges; see [3, Corollary 8.14].

Let $J_b$ be the ideal of the first copy of $\mathcal{O}_q(\mathrm{SL}_n)$ generated by $t_{ij}$ with $i < j$, and let $J_c$ be the ideal of the second copy generated by $t_{ij}$ with $i > j$. Let $I_{\mathrm{top}}$ and $I_{\mathrm{bottom}}$ be the respective bad-corner ideals in $S$.

The bad-arc normality theorem [4, Theorem 7.2, especially Equation (128)] says that each corner ideal is both the left and the right ideal generated by either one of its two oriented families of bad arcs. In our notation this gives

$$I_{\mathrm{top}} = \mathrm{span}_{\mathbb{K}}\{b_{ij} : i < j\}\, S, \qquad I_{\mathrm{bottom}} = S\, \mathrm{span}_{\mathbb{K}}\{c_{ij} : i > j\}.$$

In particular, reversed orientations introduce no further ideal.

Using (15), the first equality implies

$$I_{\mathrm{top}} \subseteq m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The reverse inclusion holds because $I_{\mathrm{top}}$ is a two-sided ideal containing the generators of $J_b$. Hence

$$I_{\mathrm{top}} = m(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n)).$$

The same argument, using the displayed right-ideal description at the lower corner, gives

$$I_{\mathrm{bottom}} = m(\mathcal{O}_q(\mathrm{SL}_n) \otimes J_c).$$

Therefore

$$I^{\mathrm{bad}} = m\big(J_b \otimes \mathcal{O}_q(\mathrm{SL}_n) + \mathcal{O}_q(\mathrm{SL}_n) \otimes J_c\big).$$

Taking the quotient of the vector-space isomorphism (15) proves (13). The injectivity of the two factor maps follows as well. □

We record two elementary consequences that will also make the eventual isomorphism explicit on all the arcs. The diagonal entries of either triangular quantum matrix commute, and

$$\prod_{a=1}^{n} b_{aa} = \prod_{a=1}^{n} c_{aa} = 1. \tag{16}$$

Thus every diagonal entry is invertible. Expanding the quantum-matrix relation on the indicated $2 \times 2$ submatrices gives, for $i < j-1$,

$$b_{ji} = \lambda^{-1} b_{j-1,j-1}^{-1} [b_{j-1,i}, b_{j,j-1}], \tag{17}$$

$$c_{ij} = \lambda^{-1} c_{j-1,j-1}^{-1} [c_{i,j-1}, c_{j-1,j}]. \tag{18}$$

Indeed, these are the relations $ad - da = \lambda bc$ for rows $j-1, j$ and columns $i, j-1$, and for rows $i, j-1$ and columns $j-1, j$, respectively. In each case the diagonal factor on the right commutes with the opposite off-diagonal factor. Consequently, each triangular algebra is generated by its diagonal and immediately adjacent off-diagonal entries.

## 3. The quantum Borel algebras and their double

3.1. **Triangular matrix realizations.** Write

$$H_- = \mathcal{U}_q(\mathfrak{b}_-), \qquad H_+ = \mathcal{U}_q(\mathfrak{b}_+).$$

**Lemma 3.1** (Borel matrix realizations)**.** *There are algebra isomorphisms*

$$\eta_- : H_- \xrightarrow{\sim} \mathcal{T}^-, \qquad \eta_+ : H_+ \xrightarrow{\sim} \mathcal{T}^+$$

*given by*

$$\eta_-(\widehat{K}_a) = b_{aa}^{-1}, \qquad \eta_-(F_i) = \lambda^{-1} b_{i+1,i+1}^{-1} b_{i+1,i}, \tag{19}$$

$$\eta_+(\widehat{K}_a) = c_{aa}, \qquad \eta_+(E_i) = -\lambda^{-1} c_{i,i+1} c_{i+1,i+1}^{-1}. \tag{20}$$

*Proof.* For $\mathfrak{gl}_n$, with independent toral generators $G_1, \dots, G_n$, these are exactly the algebra isomorphisms $\zeta_-$ and $\zeta_+$ in [1, Section 2.4]. Their inverses on adjacent entries are

$$t_{i+1,i} \longmapsto \lambda G_{i+1}^{-1} F_i, \qquad t_{i,i+1} \longmapsto -\lambda E_i G_{i+1},$$

and the diagonal entries map to $G_a^{-1}$ and $G_a$, respectively.

Impose the central relation $\prod_a G_a = 1$. On the matrix sides its image is the relation that the product of the diagonal entries is 1. For a triangular matrix this is exactly the quantum determinant-one relation. Taking these quotients gives (19)–(20). The toral lattice is

$\mathbb{Z}^n/\mathbb{Z}(1,\dots,1)$, namely the weight lattice of type $A_{n-1}$, so the resulting Borel algebras are the simply-connected ones.

Only the algebra assertion of [1, Section 2.4] is used: the coproduct convention there is opposite to (7). □

3.2. **A presentation of the generalized double.** Let

$$D = H_- \bowtie H_+$$

be the generalized quantum double for the pairing in the question. We denote its two toral copies by $\widehat{K}_a^-$ and $\widehat{K}_a^+$, and set

$$K_i^- = \widehat{K}_i^-(\widehat{K}_{i+1}^-)^{-1}, \qquad K_i^+ = \widehat{K}_i^+(\widehat{K}_{i+1}^+)^{-1}.$$

The superscript $-$ labels the negative Borel copy; it does not mean an inverse.

For clarity, we specify the double convention. The skew pairing $\tau : H_+ \otimes H_- \to \mathbb{K}$ satisfies

$$\tau(xx', y) = \tau(x, y_{(1)})\tau(x', y_{(2)}),$$
$$\tau(x, yy') = \tau(x_{(1)}, y')\tau(x_{(2)}, y),$$

and has generator values

$$\tau(\widehat{K}_a, \widehat{K}_b) = q^{1/n-\delta_{ab}},$$
$$\tau(E_i, F_j) = -\delta_{ij}\lambda^{-1}, \tag{21}$$
$$\tau(\widehat{K}_a, F_j) = \tau(E_i, \widehat{K}_b) = 0.$$

The underlying vector space of $D$ is $H_- \otimes H_+$, with multiplication identifying each factor with a subalgebra. Its exchange convention is

$$\sum \tau(x_{(1)}, y_{(1)})x_{(2)}y_{(2)} = \sum y_{(1)}x_{(1)}\tau(x_{(2)}, y_{(2)}), \qquad x \in H_+, \quad y \in H_-. \tag{22}$$

Writing this convention explicitly avoids ambiguity with opposite-double conventions.

**Lemma 3.2** (Presentation of the double)**.** *The algebra $D$ is presented by the two Borel presentations and the following mixed relations:*

$$[\widehat{K}_a^+, \widehat{K}_b^-] = 0, \tag{23}$$

$$\widehat{K}_a^+ F_i(\widehat{K}_a^+)^{-1} = q^{-\omega_{ai}}F_i, \qquad \widehat{K}_a^- E_i(\widehat{K}_a^-)^{-1} = q^{\omega_{ai}}E_i, \tag{24}$$

$$[E_i, F_j] = \delta_{ij}\frac{K_i^+ - (K_i^-)^{-1}}{\lambda}. \tag{25}$$

*Proof.* First derive the relations in $D$. For the toral terms,

$$\tau(\widehat{K}_a, K_i) = \frac{q^{1/n-\delta_{ai}}}{q^{1/n-\delta_{a,i+1}}} = q^{-\omega_{ai}}.$$

Substitution into (22) proves (23) and (24).

For $x = E_i$ and $y = F_j$, the left and right sides of (22), using (7), are

$$E_iF_j - \delta_{ij}\lambda^{-1}K_i^+, \qquad F_jE_i - \delta_{ij}\lambda^{-1}(K_j^-)^{-1},$$

respectively. This proves (25).

To see that these are a complete set of mixed relations, let $D_0$ be the algebra given by the two Borel presentations and (23)–(25). The derived relations give a surjection $D_0 \to D$.

They also allow every word in $D_0$ to be written as a sum of words with the negative Borel generators first and the positive Borel generators second. Indeed, the toral exchanges merely introduce scalars; the exchange of $E_i$ with $F_j$ has a reordered term and terms with fewer $E$'s

and $F$'s. Induction first on the number of such generators and then on the number of mixed inversions proves the assertion. Hence multiplication gives a surjection

$$H_- \otimes H_+ \longrightarrow D_0.$$

Its composition with $D_0 \to D$ is the defining vector-space isomorphism $H_- \otimes H_+ \to D$. It follows that both maps are isomorphisms, proving the presentation. □

## 4. The explicit isomorphism

Define elements of the two Borel factors, for $1 \le i < j \le n$, by

$$\mathsf{F}_{i,i+1} = F_i, \qquad \mathsf{F}_{ij} = \mathsf{F}_{i,j-1}F_{j-1} - q^{-1}F_{j-1}\mathsf{F}_{i,j-1} \qquad (j > i+1), \tag{26}$$

$$\mathsf{E}_{i,i+1} = E_i, \qquad \mathsf{E}_{ij} = E_{j-1}\mathsf{E}_{i,j-1} - q\,\mathsf{E}_{i,j-1}E_{j-1} \qquad (j > i+1). \tag{27}$$

These recursions fix all root-vector normalizations needed below.

**Theorem 4.1.** *There is an algebra isomorphism*

$$\Phi : \overline{\mathscr{S}}_n(\mathfrak{B}^*) \xrightarrow{\sim} \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$$

*given on the horizontal arcs by*

$$\Phi(b_{aa}) = (\widehat{K}_a^-)^{-1}, \qquad \Phi(c_{aa}) = \widehat{K}_a^+, \tag{28}$$

$$\Phi(b_{ji}) = \lambda(\widehat{K}_j^-)^{-1}\mathsf{F}_{ij}, \qquad \Phi(c_{ij}) = -\lambda\mathsf{E}_{ij}\widehat{K}_j^+ \qquad (i < j). \tag{29}$$

*The other off-diagonal arcs vanish by* (1).

*Equivalently, the inverse isomorphism $\Psi = \Phi^{-1}$ is determined by*

$$\Psi(\widehat{K}_a^-) = b_{aa}^{-1}, \qquad \Psi(\widehat{K}_a^+) = c_{aa}, \tag{30}$$

$$\Psi(F_i) = \lambda^{-1}b_{i+1,i+1}^{-1}b_{i+1,i}, \qquad \Psi(E_i) = -\lambda^{-1}c_{i,i+1}c_{i+1,i+1}^{-1}. \tag{31}$$

*Proof.* We first construct $\Psi$ and prove that it is an isomorphism.

For brevity put, in $A = \overline{\mathscr{S}}_n(\mathfrak{B}^*)$,

$$d_a = b_{aa}, \qquad t_a = c_{aa}, \qquad x_i = b_{i+1,i}, \qquad y_i = c_{i,i+1}.$$

The diagonal elements are invertible by (16). Lemma 3.1 gives algebra homomorphisms from the two Borel factors to $A$ with the values (30)–(31). Thus it remains to check the mixed relations in Lemma 3.2.

The internal quantum-matrix relations and (12) give

$$[d_a, d_b] = [t_a, t_b] = [d_a, t_b] = 0 \tag{32}$$

and

$$\begin{aligned} d_a x_i &= q^{\omega_{ai}} x_i d_a, \quad & t_a x_i &= q^{-\omega_{ai}} x_i t_a, \\ d_a y_i &= q^{-\omega_{ai}} y_i d_a, \quad & t_a y_i &= q^{\omega_{ai}} y_i t_a. \end{aligned} \tag{33}$$

For example, setting the first matrix entry in (12) equal to $b_{aa}$ makes both correction terms vanish by lower triangularity, and yields

$$d_a c_{kl} = q^{\delta_{al}-\delta_{ak}} c_{kl} d_a.$$

Setting the second entry equal to $c_{aa}$ gives

$$t_a b_{kl} = q^{\delta_{ak}-\delta_{al}} b_{kl} t_a.$$

These are the mixed assertions in (33); the internal ones are obtained in the same way from the single-matrix relation.

Write

$$\mathcal{F}_i = \lambda^{-1} d_{i+1}^{-1} x_i, \qquad \mathcal{E}_i = -\lambda^{-1} y_i t_{i+1}^{-1}.$$

Equations (32) and (33) immediately give

$$t_a \mathcal{F}_i t_a^{-1} = q^{-\omega_{ai}} \mathcal{F}_i, \qquad d_a^{-1} \mathcal{E}_i d_a = q^{\omega_{ai}} \mathcal{E}_i.$$

Thus the mixed toral relations are satisfied.

For the remaining mixed relation, substitute

$$(i,k;j,l) = (j+1, i; j, i+1)$$

into (12). Triangularity makes both correction terms vanish unless $i = j$, so that

$$q^{\delta_{i,j+1}} x_j y_i - q^{\delta_{j,i+1}} y_i x_j = \lambda \delta_{ij} (d_{i+1} t_i - d_i t_{i+1}). \tag{34}$$

Using (33) to move the diagonal factors gives

$$\begin{aligned} [\mathcal{E}_i, \mathcal{F}_j] &= \lambda^{-2} d_{j+1}^{-1} \left( x_j y_i - q^{\delta_{j,i+1} - \delta_{i,j+1}} y_i x_j \right) t_{i+1}^{-1} \\ &= \delta_{ij} \lambda^{-1} d_{i+1}^{-1} (d_{i+1} t_i - d_i t_{i+1}) t_{i+1}^{-1} \\ &= \delta_{ij} \frac{t_i t_{i+1}^{-1} - d_i d_{i+1}^{-1}}{\lambda}. \end{aligned} \tag{35}$$

But under the proposed map,

$$K_i^+ \longmapsto t_i t_{i+1}^{-1}, \qquad (K_i^-)^{-1} \longmapsto d_i d_{i+1}^{-1}.$$

Therefore (35) is precisely (25). Lemma 3.2 now supplies an algebra homomorphism

$$\Psi : D \longrightarrow A$$

with the required values.

We prove its bijectivity without making an unsupported presentation claim about $A$. The following diagram commutes:

$$\begin{array}{ccc} H_- \otimes_{\mathbb{K}} H_+ & \xrightarrow{m_D} & D \\ {\scriptstyle \eta_- \otimes \eta_+}\big\downarrow \cong & & \big\downarrow {\scriptstyle \Psi} \\ \mathcal{T}^- \otimes_{\mathbb{K}} \mathcal{T}^+ & \xrightarrow{m_A} & A. \end{array}$$

The top arrow is the defining vector-space isomorphism for the generalized double. The left arrow is an isomorphism by Lemma 3.1, and the bottom arrow is an isomorphism by Lemma 2.1. Consequently $\Psi$ is a vector-space isomorphism and hence an algebra isomorphism.

It remains to verify the formulas for its inverse on the non-adjacent entries. The inverse formulas on the diagonal and adjacent entries follow directly from (30)–(31). For completeness, the induction using (17) and (18) is as follows.

Put

$$m_a = (\widehat{K}_a^-)^{-1}, \qquad p_a = \widehat{K}_a^+.$$

For $i < j-1$, assume the asserted formulas at distance $j-i-1$. The toral weight relations give

$$\begin{aligned} &\lambda^{-1} m_{j-1}^{-1} [\, \lambda m_{j-1} \mathsf{F}_{i,j-1}, \lambda m_j F_{j-1} \,] \\ &\qquad = \lambda m_j \big( \mathsf{F}_{i,j-1} F_{j-1} - q^{-1} F_{j-1} \mathsf{F}_{i,j-1} \big) = \lambda m_j \mathsf{F}_{ij}. \end{aligned}$$

Here $m_j$ commutes with $\mathsf{F}_{i,j-1}$ and $F_{j-1} m_{j-1} = q^{-1} m_{j-1} F_{j-1}$. This proves the lower-triangular formula.

Similarly,

$$\lambda^{-1}p_{j-1}^{-1}[-\lambda\mathsf{E}_{i,j-1}p_{j-1}, -\lambda E_{j-1}p_j] \\ = -\lambda\big(E_{j-1}\mathsf{E}_{i,j-1} - q\mathsf{E}_{i,j-1}E_{j-1}\big)p_j = -\lambda\mathsf{E}_{ij}p_j.$$

Indeed, $p_j$ commutes with $\mathsf{E}_{i,j-1}$, $p_{j-1}E_{j-1} = qE_{j-1}p_{j-1}$, and the total weight of $\mathsf{E}_{i,j-1}E_{j-1}$ has zero coefficient at $j-1$. The latter weight statement follows by induction directly from (27). This proves the upper-triangular formula and completes the proof. □

## 5. The diagonal quotient

**Theorem 5.1.** *Let $I$ be the two-sided ideal of $\overline{\mathscr{S}}_n(\mathfrak{B}^*)$ generated by*

$$b_{aa}c_{aa} - 1, \qquad 1 \le a \le n.$$

*Then*

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I \cong \mathcal{U}_q(\mathfrak{sl}_n),$$

*where the quantum group is the simply-connected one.*

*Writing bars for classes in the quotient, the inverse isomorphism from $\mathcal{U}_q(\mathfrak{sl}_n)$ is explicitly*

$$\widehat{K}_a \longmapsto \overline{c}_{aa} = \overline{b}_{aa}^{-1}, \tag{36}$$

$$E_i \longmapsto -\lambda^{-1}\overline{c}_{i,i+1}\overline{c}_{i+1,i+1}^{-1}, \tag{37}$$

$$F_i \longmapsto \lambda^{-1}\overline{b}_{i+1,i+1}^{-1}\overline{b}_{i+1,i}. \tag{38}$$

*Proof.* By Theorem 4.1,

$$\Phi(b_{aa}c_{aa} - 1) = (\widehat{K}_a^-)^{-1}\widehat{K}_a^+ - 1.$$

Since $\widehat{K}_a^-$ is a unit, these elements generate the same ideal of $D$ as

$$\widehat{K}_a^+ - \widehat{K}_a^-, \qquad 1 \le a \le n.$$

Denote that ideal by $J$. Hence $\Phi(I) = J$ and

$$\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I \cong D/J. \tag{39}$$

There is a surjective homomorphism

$$\pi : D \longrightarrow \mathcal{U}_q(\mathfrak{sl}_n)$$

sending both $\widehat{K}_a^-$ and $\widehat{K}_a^+$ to $\widehat{K}_a$ and sending $E_i, F_i$ to the identically named generators. To check it, use Lemma 3.2: the only new commutator becomes

$$[E_i, F_j] = \delta_{ij}\frac{K_i - K_i^{-1}}{\lambda},$$

and all the other relations become (2)–(6). Clearly $J \subseteq \ker\pi$.

Conversely, in $D/J$ the two toral copies agree. Their common images together with the images of $E_i, F_i$ satisfy the entire presentation (2)–(6). They therefore define a homomorphism

$$\mathcal{U}_q(\mathfrak{sl}_n) \longrightarrow D/J.$$

This homomorphism and the map induced by $\pi$ are inverse on every generator. Consequently

$$D/J \cong \mathcal{U}_q(\mathfrak{sl}_n) \qquad \text{and} \qquad \ker\pi = J.$$

Together with (39), this proves the claimed quotient, including the assertion that no further relations have been introduced. Formulas (36)–(38) are the specializations of (30)–(31). □

*Remark* 5.2. The elements $b_{aa}c_{aa}$ are central. Under $\Phi$ they become $(\widehat{K}_a^-)^{-1}\widehat{K}_a^+$, and the two toral copies have identical conjugation actions on every $E_i$ and $F_i$. Furthermore,

$$\prod_{a=1}^{n} b_{aa}c_{aa} = 1.$$

Thus the quotient identifies the two full weight-lattice tori; there are only $n-1$ independent diagonal identifications. This is also why the conclusion is the simply-connected quantum group rather than its root-lattice form.

*Remark* 5.3 (Parameter qualification). The factors $\lambda^{-1}$ are essential to the identification with the usual Chevalley generators. At $q = \pm 1$ the skew pairing displayed in the question itself is undefined. Statements at such parameters require an integral-form reformulation and are not obtained by substituting into the formulas above.

## References


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